SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA04
COURSE NAME: Operating Systems

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments.*

QUESTIONS: 5 **Total Marks:100**

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I VAIBHAVA LAKSHMI K (192511310) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Vaibhavi. K

Annexure A

Short Answer Test

Question 1 (5 Marks)
Suppose a modern Operating System supports multiple users and processes with shared resources like CPU, memory, and I/O devices.
(a) Classify different types of Operating Systems suitable for this scenario and justify your choice.
(b) Draw the structure of an Operating System showing how resources are managed.
(c) Write a sequence of system calls required to create and execute a new process in such an environment. **(BL2)**

Question 2 (5 Marks)
Consider an Operating System managing multiple files and devices for concurrent users.
(a) Categorize system calls used for file and device management.
(b) Illustrate how the OS handles file operations using a layered structure diagram.
(c) Write a pseudo-code or C-based system call sequence to open, read, and close a file. **(BL3)**

Question 3 (5 Marks)
Assume a real-time Operating System used in an embedded system controlling industrial machinery.
(a) Identify the type of OS and explain why it is suitable for real-time applications.
(b) Draw a diagram representing the interaction between hardware, kernel, and user processes.
(c) List and explain system calls required for process synchronization in this system. **(BL2)**

Question 4 (5 Marks)
A distributed computing system consists of multiple interconnected computers sharing resources over a network.
(a) Classify the type of Operating System used and compare it with centralized OS.
(b) Draw the architecture of a distributed OS showing communication between nodes.
(c) Write the sequence of system calls involved in inter-process communication (IPC). **(BL3)**

Question 5 (5 Marks)
Consider a scenario where an Operating System must manage process scheduling, memory allocation, and device communication efficiently.
(a) Explain the major functions of an OS in handling these tasks.
(b) Draw a block diagram representing OS components and their interactions.
(c) Write system call examples for memory allocation and device management. **(BL3)**

RUBRICS FOR CALCULATION

Criteria	Weightage (Marks)	Level 4 – Exemplary (Full Marks)	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	35 Marks	Demonstrates clear, complete, and in-depth understanding of OS concepts (types, structure, system calls)	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answers	25 Marks	All answers are correct, precise, and relevant	Mostly correct with minor errors	Partially correct with noticeable errors	Mostly incorrect or irrelevant
Application of Knowledge	15 Marks	Effectively applies concepts to scenarios/questions	Applies with minor mistakes	Limited or partial application	No meaningful application
Completeness of Response	15 Marks	All parts of each question are fully answered	Most parts covered with small omissions	Some parts missing	Incomplete answers
Clarity & Presentation	10 Marks	Clear, well-structured, and neatly presented answers	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

DATE: 7.10.2026.

NAME: VAIBHAVA LAKSHMI.K
REG. NO: 192511310
COURSE: Operating Systems
COURSE CODE: CSA0414.

CO1 ASSESSMENT TOOL-2.

1) **Types of OS:**

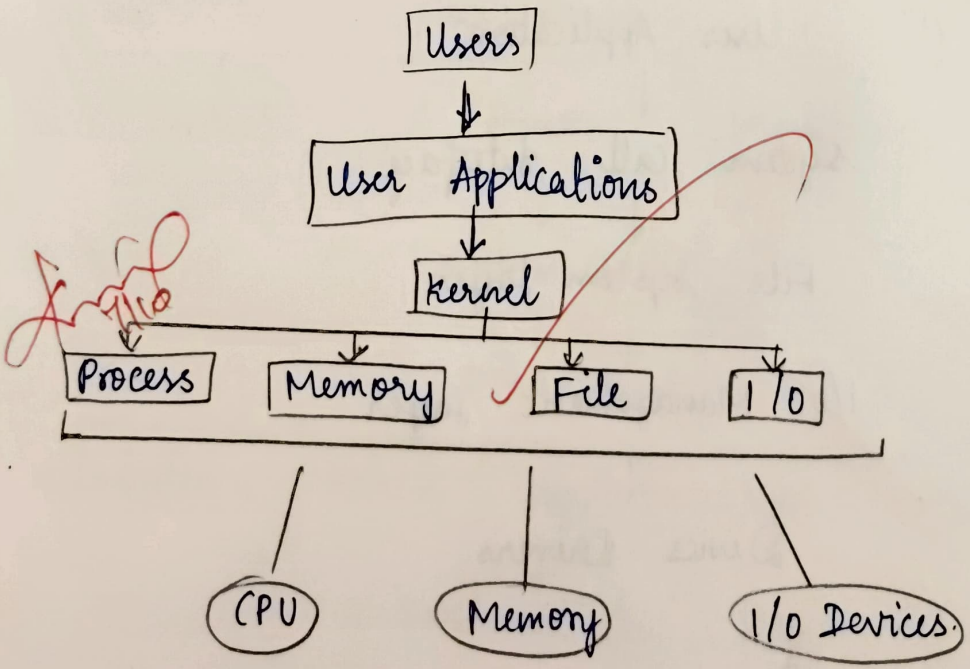
⇒ Multi-user OS: Allows several users to access system resources simultaneously while maintaining security.

⇒ Multi-tasking OS: Allows multiple processes to execute concurrently by sharing the CPU.

⇒ Multiprocessing OS: Supports multiple CPUs

⇒ Time-sharing OS: Allocates CPU time slices.

⇒ Resource Management:



c) System Calls:

- fork() → Process is created
- exec() → New program is loaded
- wait() → Allows parents to wait
- exit() → Terminates the process.

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2) a) Categorise system calls:

- Open() - opens a file
- read() - Reads data
- write() - Writes data to a file
- close() - closes an opened file
- lseek() - changes file position.

b) Layered Structure for file operations:

User Application
↓
System call Interface
↓
File System layer
↓
I/O Management layer
↓
Device Drivers
↓
Hardware / Storage.

c)
```
int fd;
char data[100];
fd = open("file.txt", O_RDONLY);
read(fd, data, 100);
close(fd);
```

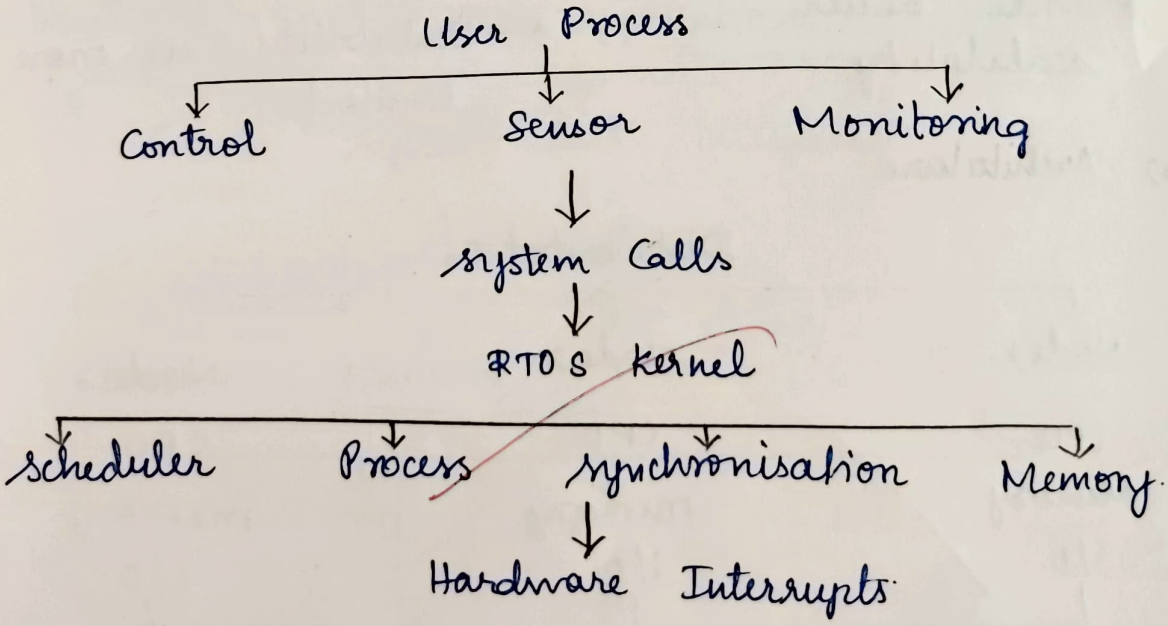
3)

a) Type of OS

=> Real-Time Operating System (RTOS).

* Provides fast response to sensor
* Uses priority-based scheduling.
* Provides predictable timing
* Improves reliability & safety

b) Interaction between Hardware, Kernel.



c) System Calls:

- sem_wait()
- sem_post()
- pthread_mutex_lock()
- pthread_cond_wait()

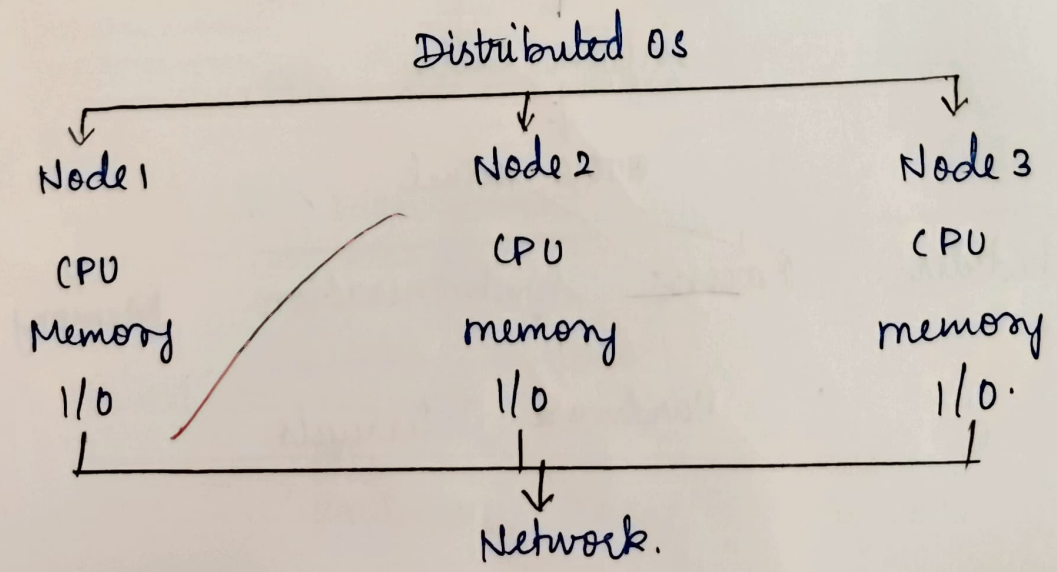
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d) Distributed OS:

a)

Distributed OS	Centralised OS
* Uses multiple computers	* Mainly manages one central computer
* Resources are distributed.	* It is Concentrated in one system
* Support sharing across nodes.	* Sharing is within the central system
* Provide better scalability	* Scalability is more limited.

b) Architecture:



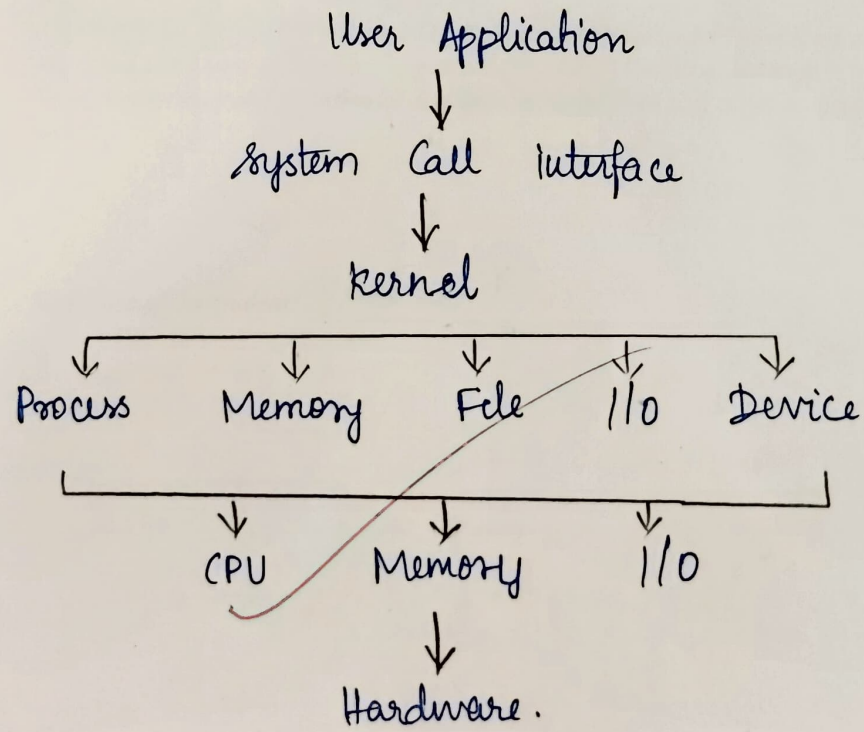
c)
- pipe()
- fork()
- write()
- read()
- close()

5)

a) Major function includes:

1) Process Management
2) Memory Management
3) Device Management
4) File Management
5) Resource Management.

b) Block diagram.



c) System Calls:

- mmap()
- munmap()
- open()
- read()
- ioctl()
- close()